

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 13 — Capital Budgeting: NPV, IRR, and Payback

Drill Set

Required: 1–5. Optional: 6–8. Complete the required items after instruction by **April 18, 2027, 11:59 p.m. Eastern**. Work individually by hand, check the key, and correct errors. Graded for completion; optional items are excluded from quizzes and the final.

1. (*Payback, even cash flows.*) A project costs \$240,000 and generates \$60,000 per year. Compute the payback period assuming cash arrives uniformly within each year.
2. (*Payback, uneven cash flows.*) A project costs \$100,000 and generates \$30,000, \$40,000, \$50,000, and \$60,000 in years 1 through 4. Compute the payback period assuming cash arrives uniformly within each year.
3. (*Present value of a single sum.*) A project returns a single cash flow of \$50,000 at the end of year 3. At a 10% discount rate, compute its present value.
4. (*NPV, annuity.*) A project costs \$75,000 and generates \$25,000 per year for four years. The discount rate is 10%. Compute the net present value.
5. (*NPV, uneven cash flows.*) A project costs \$60,000 and generates \$20,000, \$30,000, and \$25,000 in years 1 through 3. The discount rate is 8%. Compute the net present value.
6. (*Optional: IRR decision rule.*) Assume an initial outflow followed only by positive inflows. A project has an internal rate of return of 9% and the firm's cost of capital is 11%. Should the project be accepted? At the cost of capital, is the project's NPV positive, negative, or zero?
7. (*IRR, even cash flows.*) A project costs \$79,247 and returns \$25,000 per year for four years. Compute the annuity factor (cost $\div$ annual cash flow), then identify the IRR given that the four-year annuity factor equals 3.1699 at 10%.
8. (*Profitability index.*) A project requires an initial investment of \$50,000 and the present value of its future cash flows is \$58,000. Compute the profitability index and state the decision.

Supplied factors: 10%, year 3: 0.751315; 10%, four-year annuity: 3.169865. At 8%, years 1–3: 0.925926, 0.857339, 0.793832. Round final dollars; small factor-rounding differences are acceptable.

Self-Check Answers

Required answers: 1–5. Other answers support optional practice.

1. **Payback, even.** $= 240,000 \div 60,000 = 4.0$ years.
2. **Payback, uneven.** Cumulative cash flow: year 1 = 30,000; year 2 = 70,000; year 3 = 120,000. The \$100,000 cost is recovered during year 3: after year 2, \$30,000 remains, and year 3 brings \$50,000, so $30,000 \div 50,000 = 0.6$. Payback = **2.6 years** under uniform intra-year receipts; if receipts arrive only at year-end, recovery occurs at the end of year 3.
3. **PV of a single sum.** $= 50,000 \div 1.10^3 = 50,000 \div 1.331 = \mathbf{\$37,566}$.
4. **NPV, annuity.** Annuity factor (10%, 4 years) = 3.1699. PV of inflows $= 25,000 \times 3.1699 = \$79,247$. NPV $= 79,247 - 75,000 = \mathbf{\$4,247}$.
5. **NPV, uneven.** PVs at 8%: year 1 $= 20,000 \div 1.08 = 18,519$; year 2 $= 30,000 \div 1.08^2 = 25,720$; year 3 $= 25,000 \div 1.08^3 = 19,846$. Sum = 64,084. NPV $= 64,084 - 60,000 = \mathbf{\$4,084}$.
6. **IRR rule.** For the conventional cash-flow pattern specified, **reject** — the IRR (9%) is below the cost of capital (11%). At an 11% discount rate the NPV is **negative** (NPV equals zero only at the 9% IRR, and falls as the discount rate rises).
7. **IRR, even.** Annuity factor $= 79,247 \div 25,000 = 3.1699$. Because that equals the four-year annuity factor at 10%, the IRR = **10%**.
8. **Profitability index.** $= \text{PV of future cash flows} \div \text{initial investment} = 58,000 \div 50,000 = \mathbf{1.16}$. Since $PI > 1.0$, **accept** the project.